Lecture 3
Lifecycle Management of Medical Devices

End-to-End Medical Device Management

Lesson overflow

- Understand the end-to-end life cycle of medical devices from need identification to final disposal.
- Explore maintenance strategies, including preventive maintenance, repairs, safety checks, and CMMS.
- Learn how hospitals evaluate, select, acquire, and install new medical technologies.
- Recognize replacement and de-installation processes, including end-of-life planning, data removal, and safe disposition.

Technology support life cycle

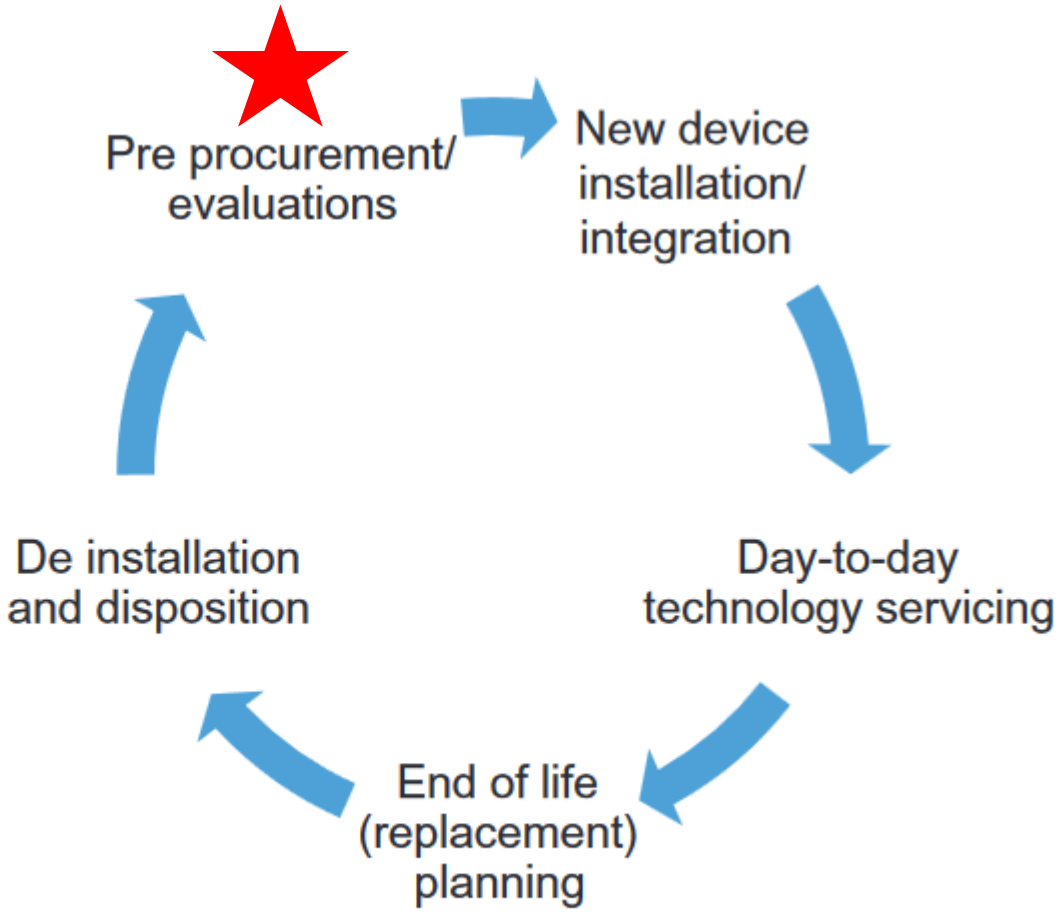


Figure 2.1 Phases of the technology support life cycle.

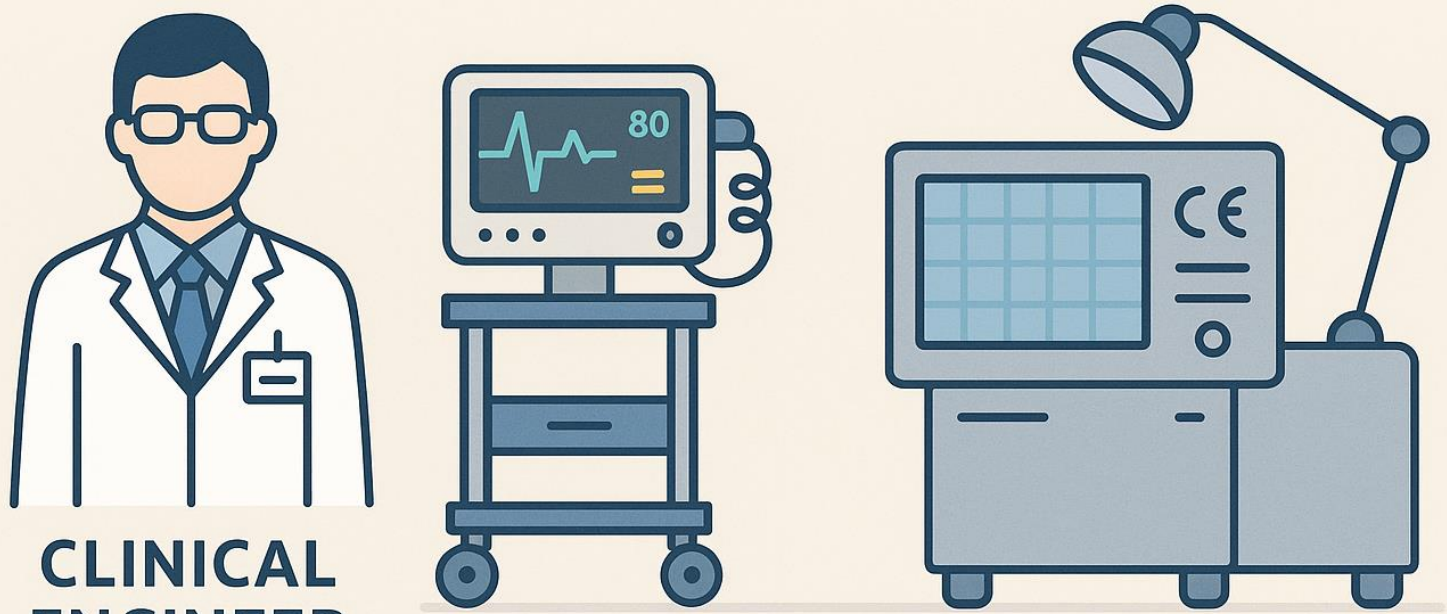
Procurement (General Definition)

- **Procurement** is the organized process of identifying needs, sourcing suppliers, requesting proposals/quotations, evaluating offers, and purchasing goods or services in a cost-effective, transparent, and fair way.

  → **It includes:**

  planning → tendering → contracting → delivery → payment.

MEDICAL DEVICE PROCUREMENT



CLINICAL ENGINEER

CLINICAL REQUIREMENTS

LIFE CYCLE COST

REGULATIONS

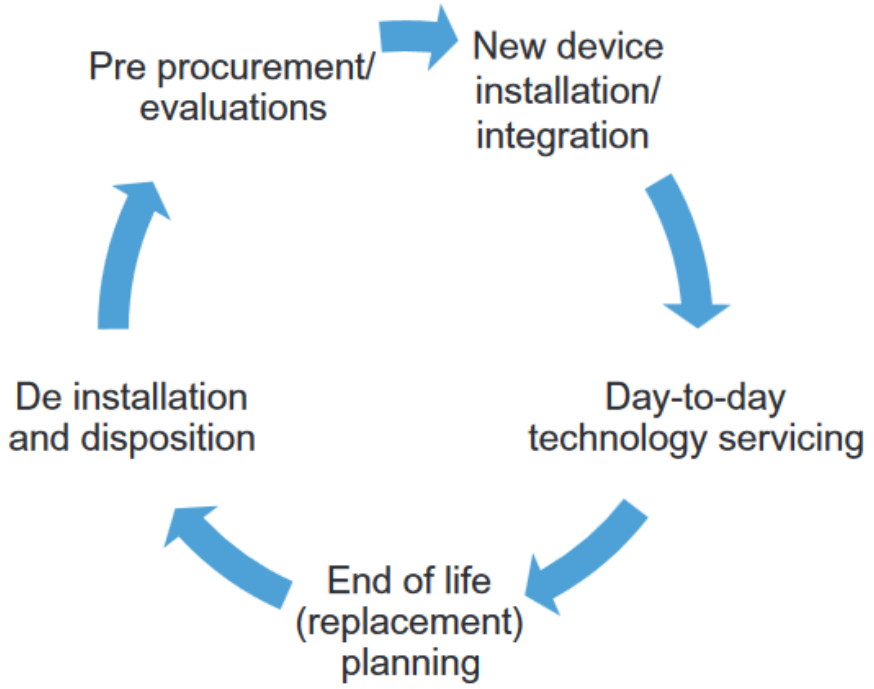
CRITICALITY

Procurement in the Medical Field (Healthcare Procurement)

- In healthcare, procurement is the specialized process of acquiring medical equipment, consumables, devices, and services while ensuring patient safety, regulatory compliance (CE/FDA), quality standards, warranty, maintenance plans, and training.
- It focuses on:
    - **Clinical requirements** (safety, performance, compatibility)
    - **Lifecycle cost** (purchase + maintenance + spare parts)
    - **Regulations** (ISO 13485, MDR, MoH rules … etc.)
    - **Criticality** (impact on patient care)
- Healthcare procurement is more complex because decisions affect patient outcomes, hospital operations, and compliance with international standards.

Technology support life cycle

- Clinical engineers manage medical technology throughout its life cycle.
- Life cycle begins before acquisition and ends with removal/disposition.
- Ensures:
  - Safe and effective technology use
  - Cost-effective management
  - Alignment with clinical needs.



Pre-Acquisition Phase

- Technology need identified by hospital/clinic staff.
- Requirements team/group ideally formed:
  - Includes key stakeholders
  - Supports multifaceted decision-making.
- Pre-procurement evaluation/assessment:
  - Determines which specific devices meet requirements.
  - Considers technical, clinical, and financial factors.

Acquisition & Implementation

- Device selection → purchase decision.
- Implementation tasks include:
  - Device installation & integration into hospital systems
  - Staff training (operation + support).
- Goal: smooth transition from procurement to clinical use.

Operational Support Phase

- Day-to-day support by clinical engineering department.
- Activities:
  - Preventive maintenance → routine inspections, calibration, servicing.
  - Reactive maintenance → repair when failures occur.
  - Incident investigations when malfunctions are observed.
  - Managing recalls & safety alerts from manufacturers/regulatory bodies.

End-of-Life & Replacement

- Device reaches end of useful life when:
  - Support/maintenance costs become too high
  - Newer technology supersedes current device.
  - Clinical engineering responsibilities:
  - Plan for replacement or removal
  - De-installation of device
  - Proper disposition (recycling, donation, disposal).

Medical Device Inventory: Importance

- Accurate inventory is critical to support devices across their life cycle.
- Joint Commission* requires every medical device to be inventoried.

*https://www.jointcommission.org/en

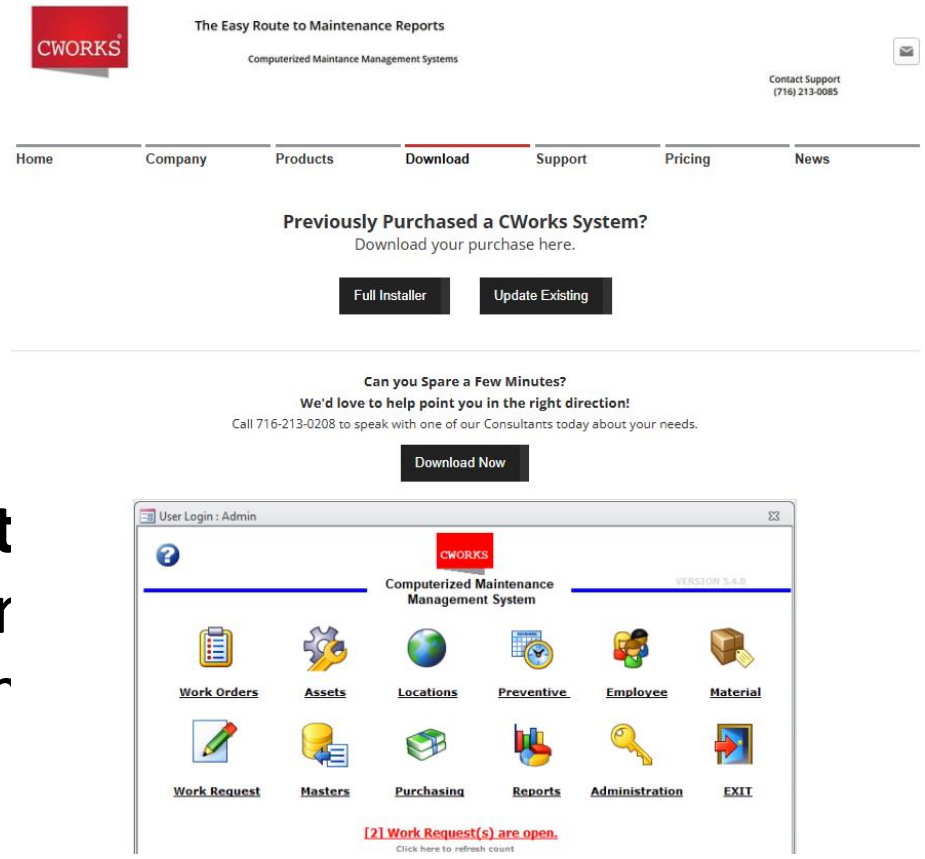


Use of CMMS (Computerized Maintenance Management System)

- Used by nearly all healthcare technology departments.
- **Tracks:**
  1. Device records (serial/model numbers, asset tag)
  2. Work orders and repair histories
  3. Spare parts and service needs
  4. Detailed history for each device.

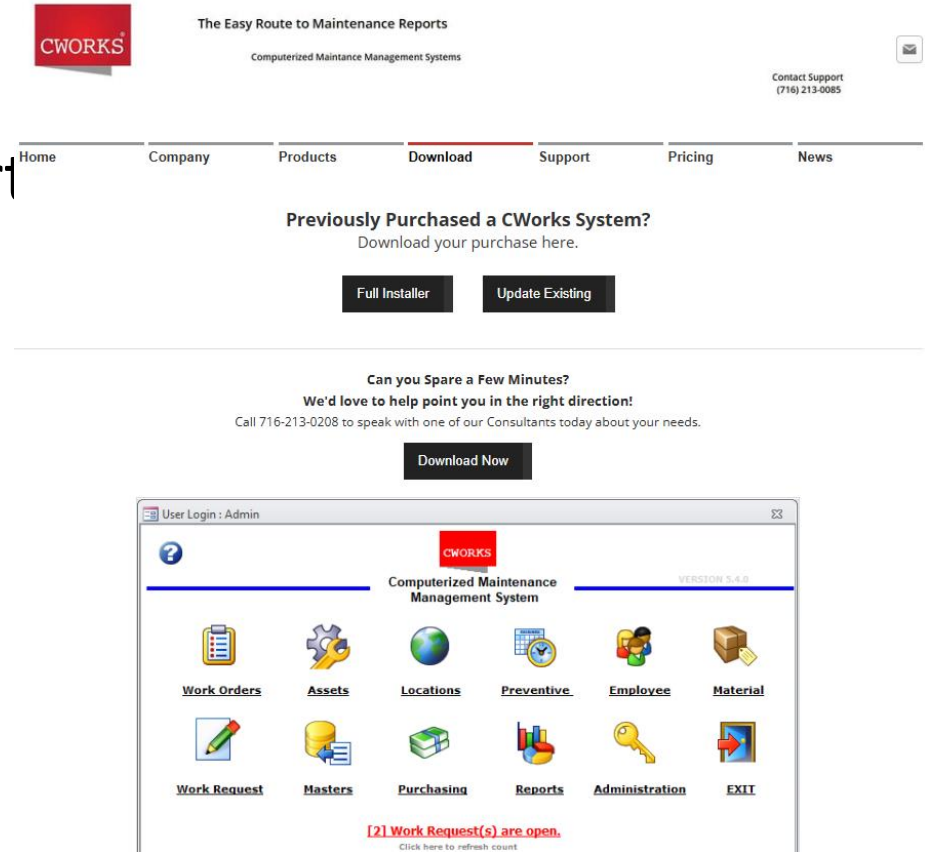
CWorks System - CMMS

- **Centralized asset & maintenance control:** Tracks all equipment details ; location, work history, depreciation, preventive and corrective actions.
- **Efficient work order management** Supports automatic & manual wor orders from request to completior improving planning and reducing downtime.



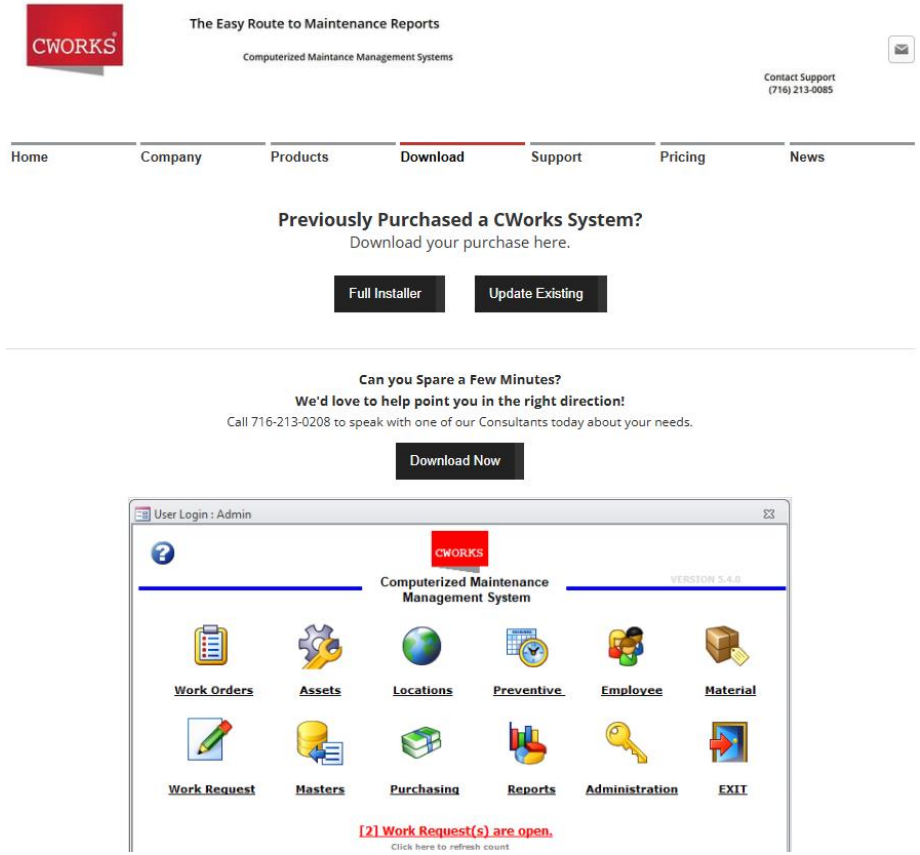
CWorks System - CMMS

- **Full inventory & purchasing management:** Monitors spare part usage, costs, stock levels, and simplifies purchasing/replenishment.
- **Reduces maintenance costs & increases uptime:** Designed to lower operational costs through strong preventive maintenance workflows.



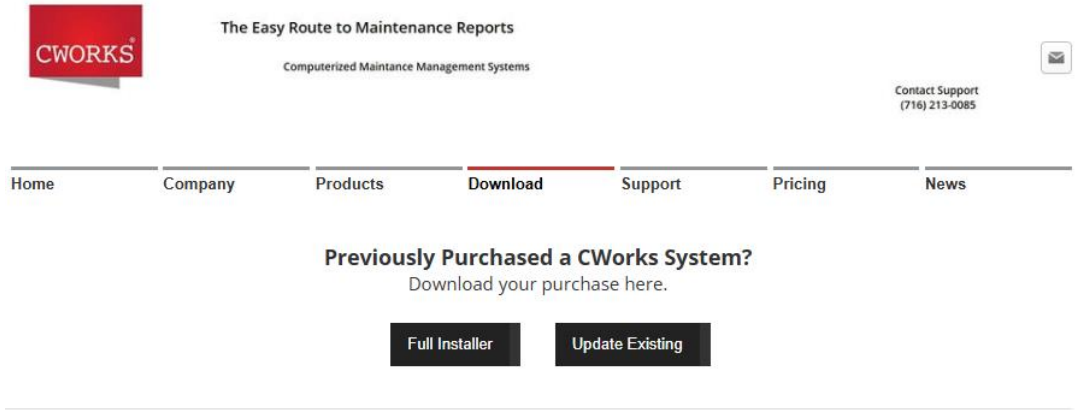
CWorks System - CMMS

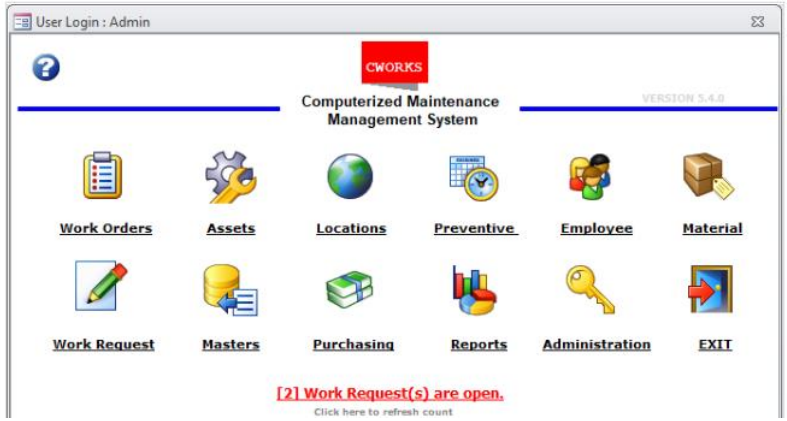
- **Fast setup & strong support:** Easy to deploy (e.g., 2,000+ assets in 2 weeks) with responsive, customizable customer support.
- **Supports medical-specific workflows** such as calibration reminders, safety checks, risk ranking, and compliance documentation.



CWorks System - CMMS

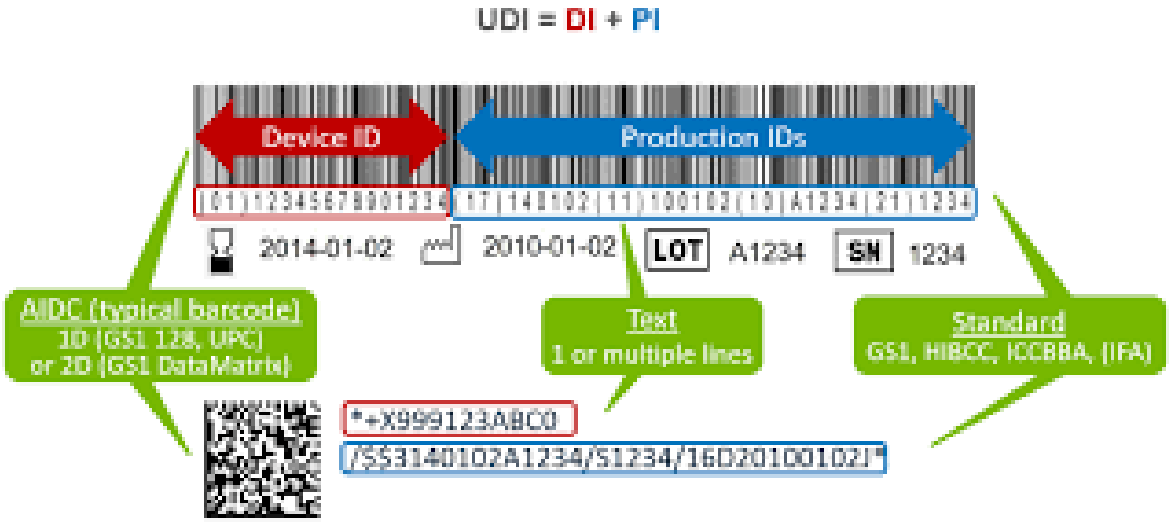
- Although CWorks is a general-purpose CMMS, we use it in our context as a *Medical CMMS* for hospital/clinical engineering asset management.
- http://www.cworkssystems.com/download.html

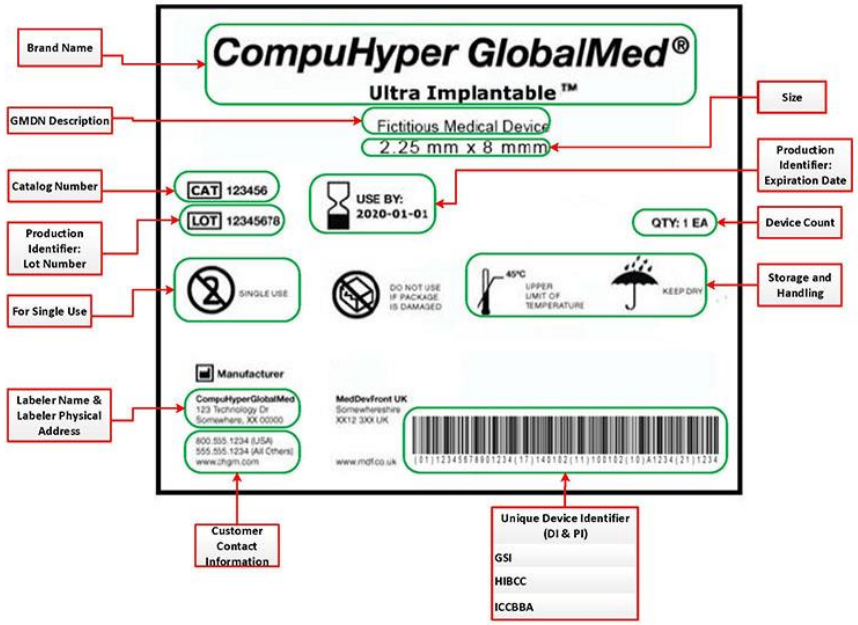




Device Identification

- Each device must have a unique tag (sticker or metal plate).
- Linked to inventory software that stores device-specific information.





Benefits of Computerized Inventory

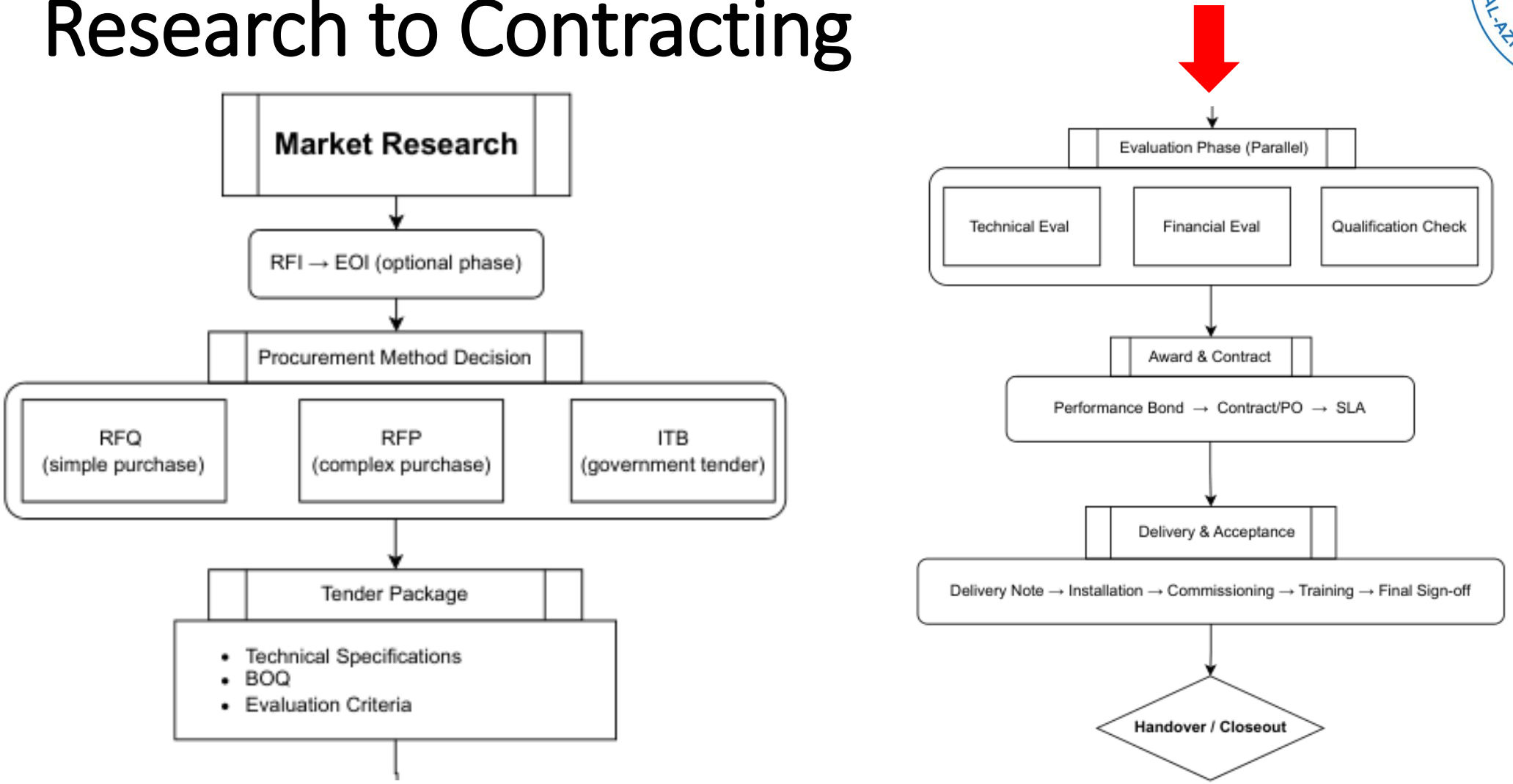
- Enables data analytics:
  - Aggregated repair data by device type/category.
  - Multi-site analytics for large health systems.
  - Predicts repair needs and interventions.
  - Ensures optimal performance and availability of technology.

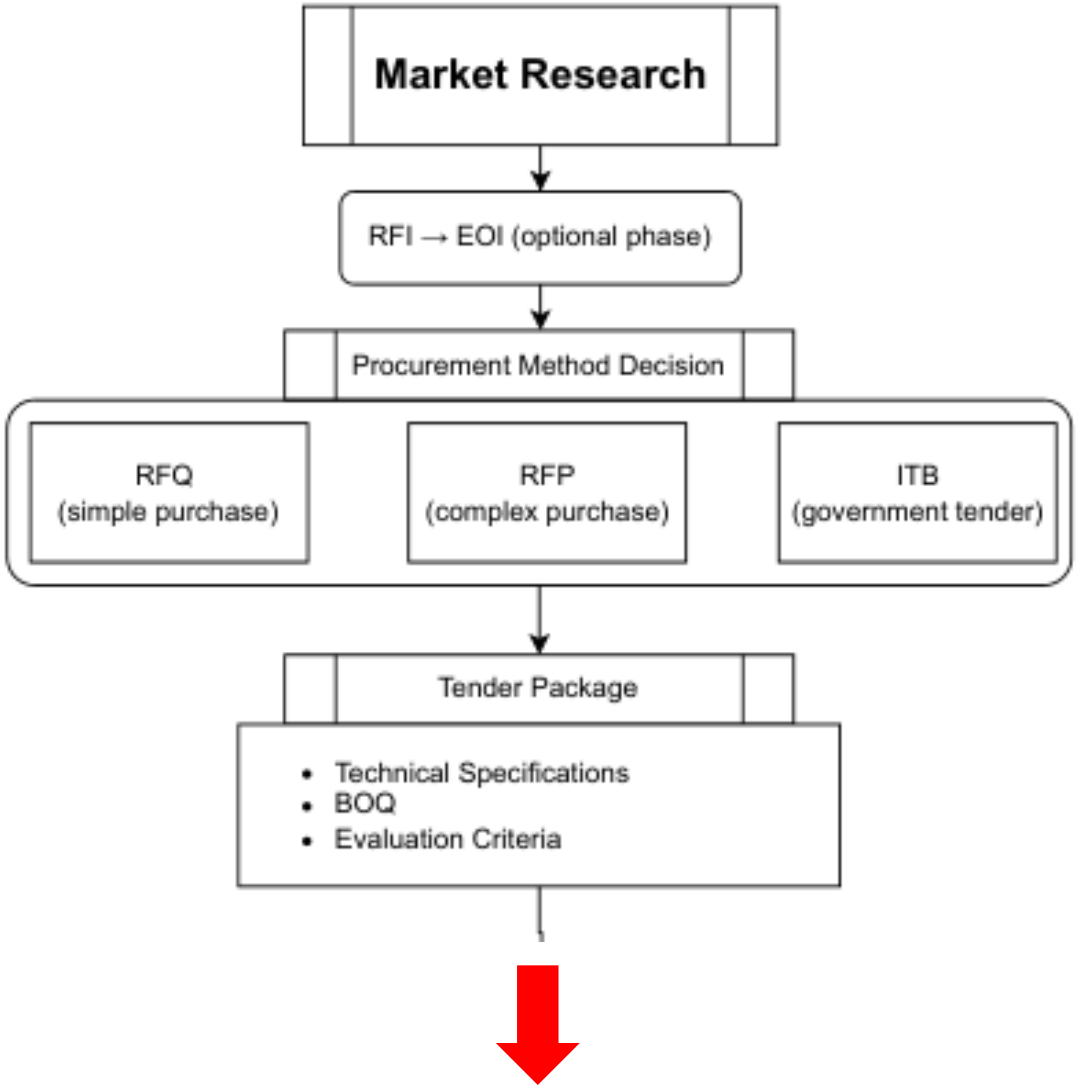


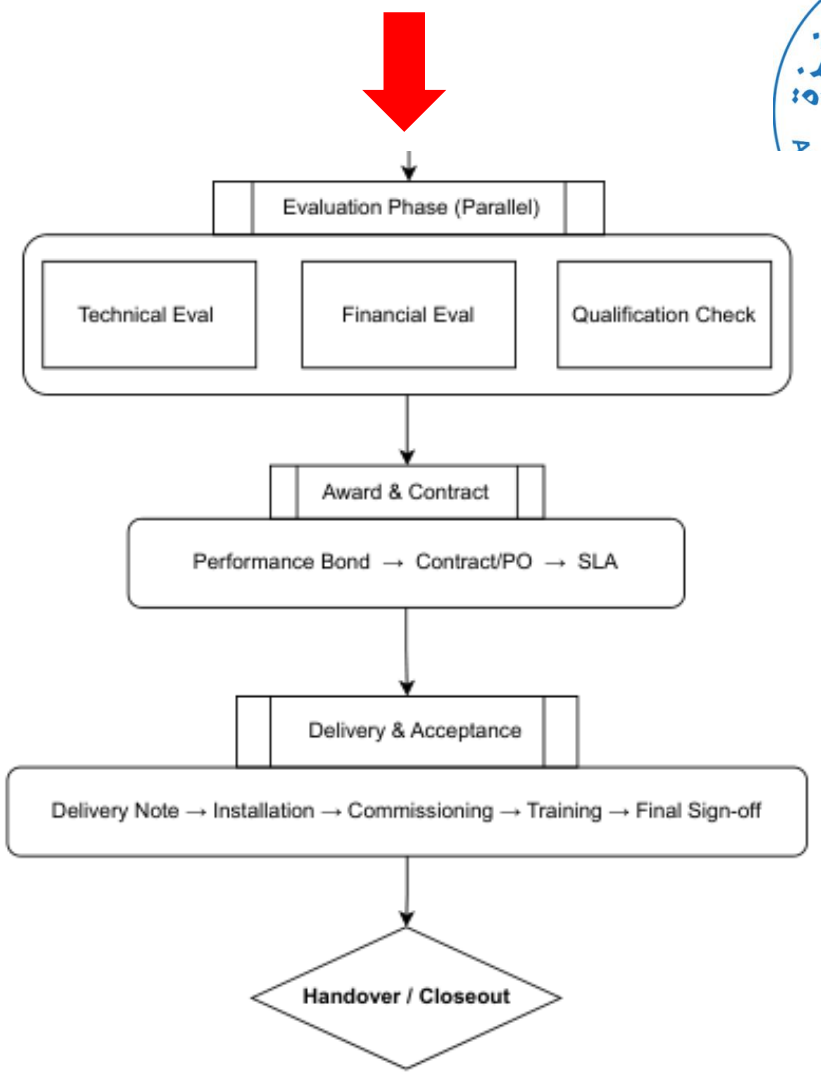
Pre-Procurement Evaluation & Selection

- Purpose: compare devices to meet clinical, technical, and financial needs.
- Selection process:
  - Some sites → individual clinician decisions.
  - Others → requirements team (engineers, staff, stakeholders).

Procurement Documents: From Market Research to Contracting




Market Research
RFI → EOI (optional phase)
Procurement Method Decision
RFQ
(simple purchase)
RFP
(complex purchase)
ITB
(government tender)
Tender Package
Technical Specifications
BOQ
Evaluation Criteria


Evaluation Phase (Parallel)
Technical Eval
Financial Eval
Qualification Check
Award & Contract
Performance Bond → Contract/PO → SLA
Delivery & Acceptance
Delivery Note → Installation → Commissioning → Training → Final Sign-off
Handover / Closeout

1. RFI – Request for Information

- **Purpose:** Market research only , to understand the market, specifications, suppliers, and price ranges.
- **When used:**
  - Before preparing the tender.
  - When you don’t yet know specs, vendors, or prices.
- **Output:**
  - List of potential suppliers
  - Technical understanding
  - Price benchmarks
  - Specification recommendations

2. EOI – Expression of Interest

- **Purpose:** To identify suppliers who are qualified and willing to participate in the future tender.
- **When used:**
  - After RFI, before tender.
  - When you need to shortlist suppliers.
- **What you collect from vendors:**
  - Company profile
  - Certifications
  - Experience
  - Legal documents
  - Registration details

3.RFQ – Request for Quotation

- **Purpose:** Used for small/medium purchases.

  Simple pricing request , no complex evaluation.
- **When used:**
  - When specifications are clear
  - For small equipment or consumables
  - For simple price comparison

4. RFP – Request for Proposal

- **Purpose:** Used for complex projects where you need:
  - Technical proposal
  - Solution design
  - Methodology
  - Experience
  - Service plan
  - Pricing
- **When used:**
  - For large and complex needs
  - When technical scoring is required
  - When each vendor may propose a different solution

5. ITB / IFB – Invitation to Bid (Tender)

- **Purpose:** The official tender document.
  - Price is the main evaluation criterion.
- **When used:**
  - When specifications are fixed
  - When you want the lowest compliant bidder
  - Standard for government tenders

6. BOQ – Bill of Quantities

- **Purpose:**
  - Detailed quantity and item list for pricing.
- **When used:**
  - Attached with ITB/RFP
  - Vendors fill prices in the BOQ

7. Technical Specifications Document

- **Purpose:** Defines the exact technical requirements.
- **When used:**
  - Attached to RFP/ITB
  - Used for technical evaluation

8. Evaluation Criteria Document

- **Purpose:** Shows how vendors will be scored.
- **Includes:**
  - Technical score
  - Financial score
  - Past performance
  - Delivery time
  - Service & warranty

9. Contract / Purchase Order

- **Purpose:** Final legal document with the winning vendor.
- **When used:**
  - After evaluation and award
  - Defines responsibilities, payments, timelines

10. SLA – Service Level Agreement

- **Purpose:** Agreement on service, maintenance, uptime, response time.
- **When used:**
  - For medical devices
  - For service/maintenance contracts

11. Performance Bond / Bank Guarantee

- **Purpose:** Guarantees vendor performance (typically 10% of contract value).
- **When used:**
  - Before contract signing
  - Required in government and UN/WHO tenders

12. Delivery & Acceptance Documents

- **Purpose:** Confirm delivery, installation, testing, and acceptance.
- **Includes:**
  - Delivery note
  - Inspection report
  - Commissioning report
  - User training certificates

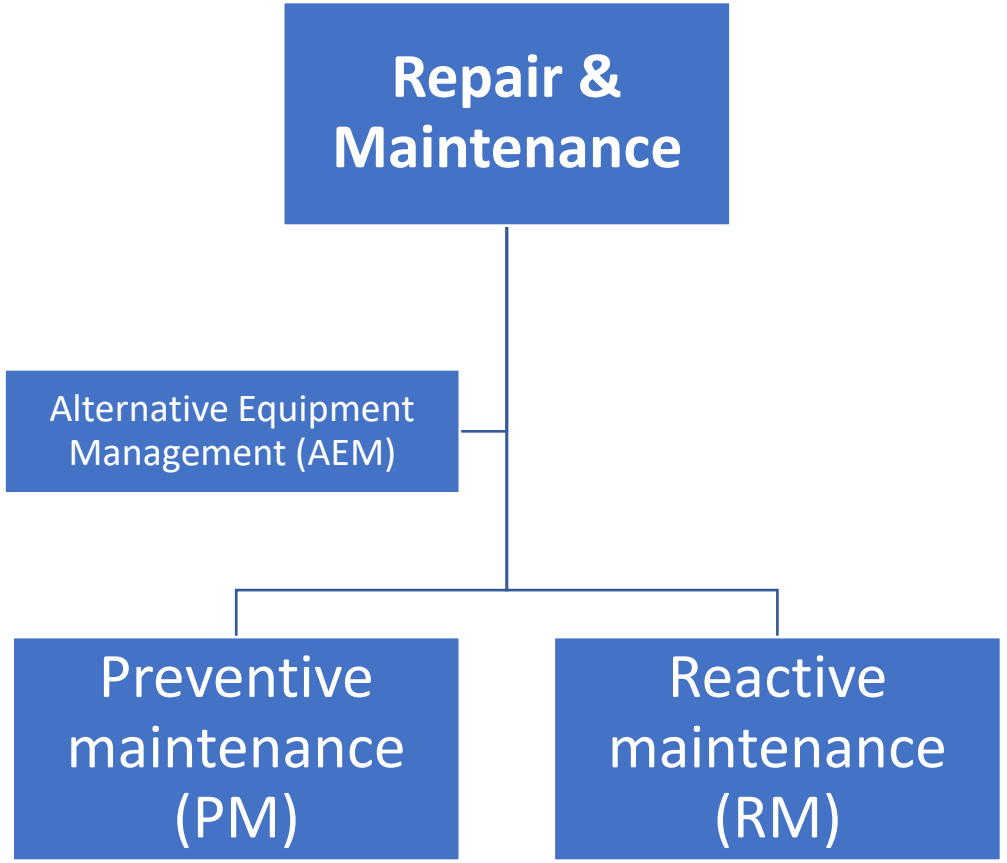
13. Final Project Handover File

- **Purpose:** Full record of all documents after project completion.



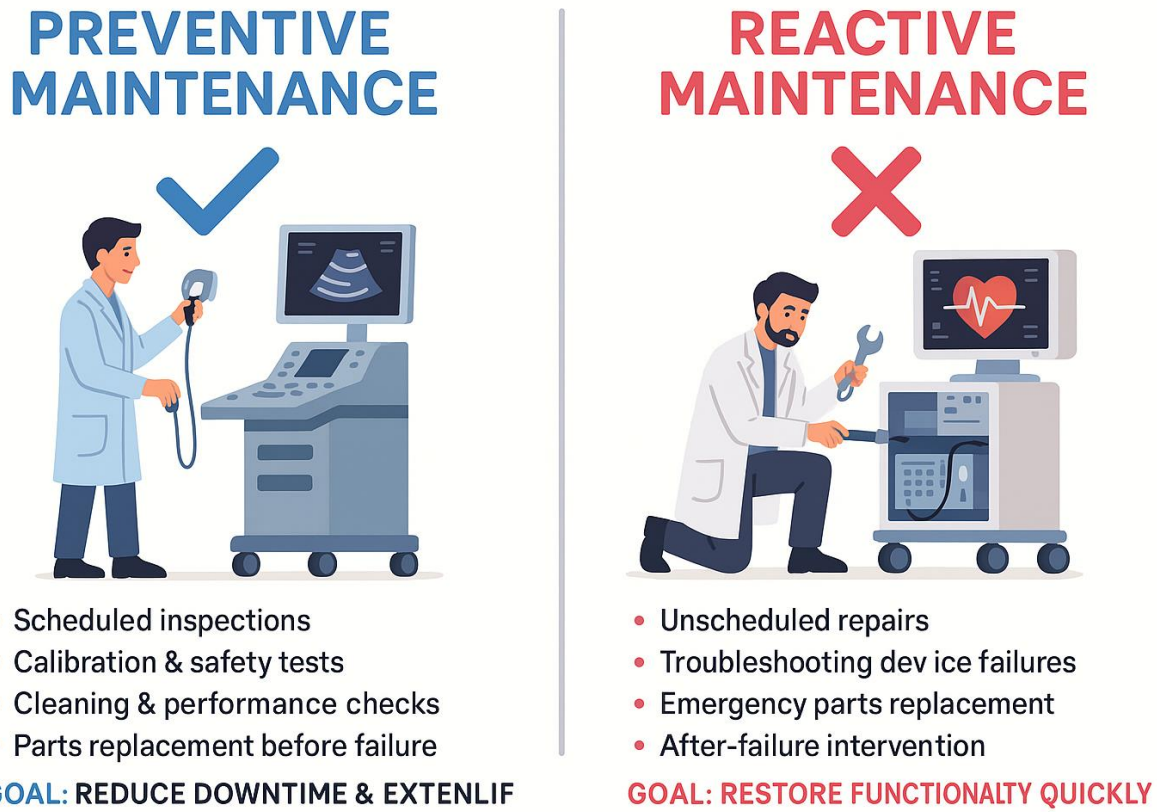
Repair & Maintenance Overview

- Two types:
  - Preventive maintenance (scheduled, proactive).
  - Reactive maintenance (unscheduled, repair after failure).
- Both are documented in device history files by Clinical Engineering.



Preventive Maintenance (PM)

- Periodic inspections, calibration, performance validation.
- Based on manufacturer manuals (min. annually).
- Includes all essential electrical safety checks, such as earth leakage, touch leakage, patient leakage, and patient auxiliary current tests.

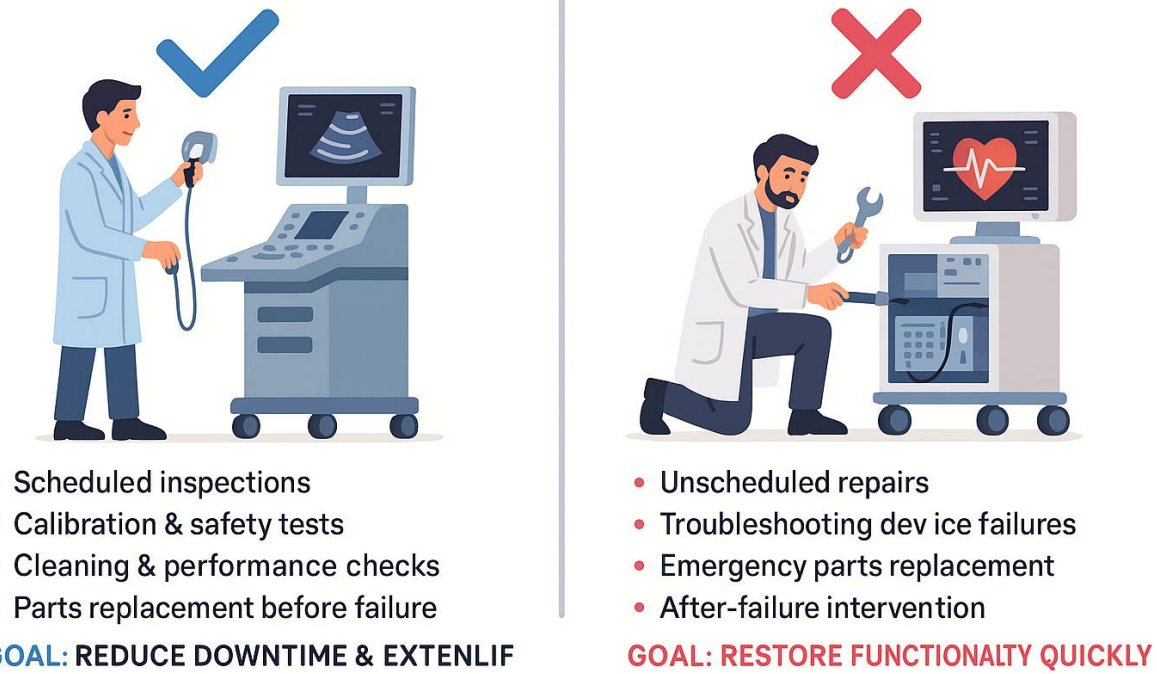


Alternative Equipment Management (AEM)

- Allows deviations from manufacturer's PM schedule.
- Based on: repair records, peer-reviewed sources, organizational experience.
- Criteria for AEM eligibility:
  - Use & risk of harm
  - Consequences of failure
  - Backup availability
  - 2+ years incident history
  - Manufacturer requirements
  - Some devices cannot be in AEM (e.g., lasers, imaging, new equipment).

Reactive Maintenance

- Unscheduled repairs → troubleshoot, replace parts, clean, calibrate.
- Critical because downtime can delay/cancel procedures.
- Strategy: “Corrective-only maintenance” may be applied to low-risk devices.

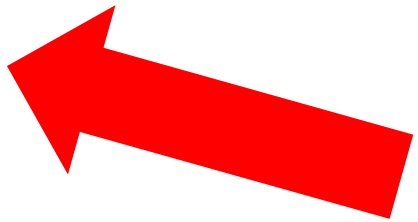


Repair Strategies & Contracts

- In-house only (full support by hospital technicians).
- In-house first, vendor if needed (flexible).
- PM-only contract (vendor does preventive tasks).
- Time & materials contract (vendor bills per service/parts).
- Contracts tied to disposables (requires supply purchase).
- Full-service contract (vendor covers everything).
- Depot repair (send devices off-site, requires backups).

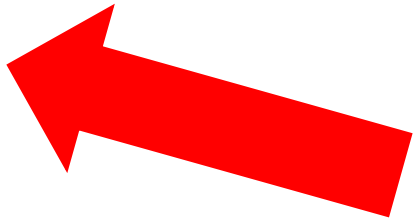
Key Challenges in Maintenance

- “No problem found” → wastes time, requires better user-technician communication.
- Parts availability & costs → need vendor comparison, group buying to reduce costs.
- Recalls & alerts → require updates (hardware/software), documentation in history.
- Patient incidents → must be investigated and documented in device file.



Device Retirement Planning

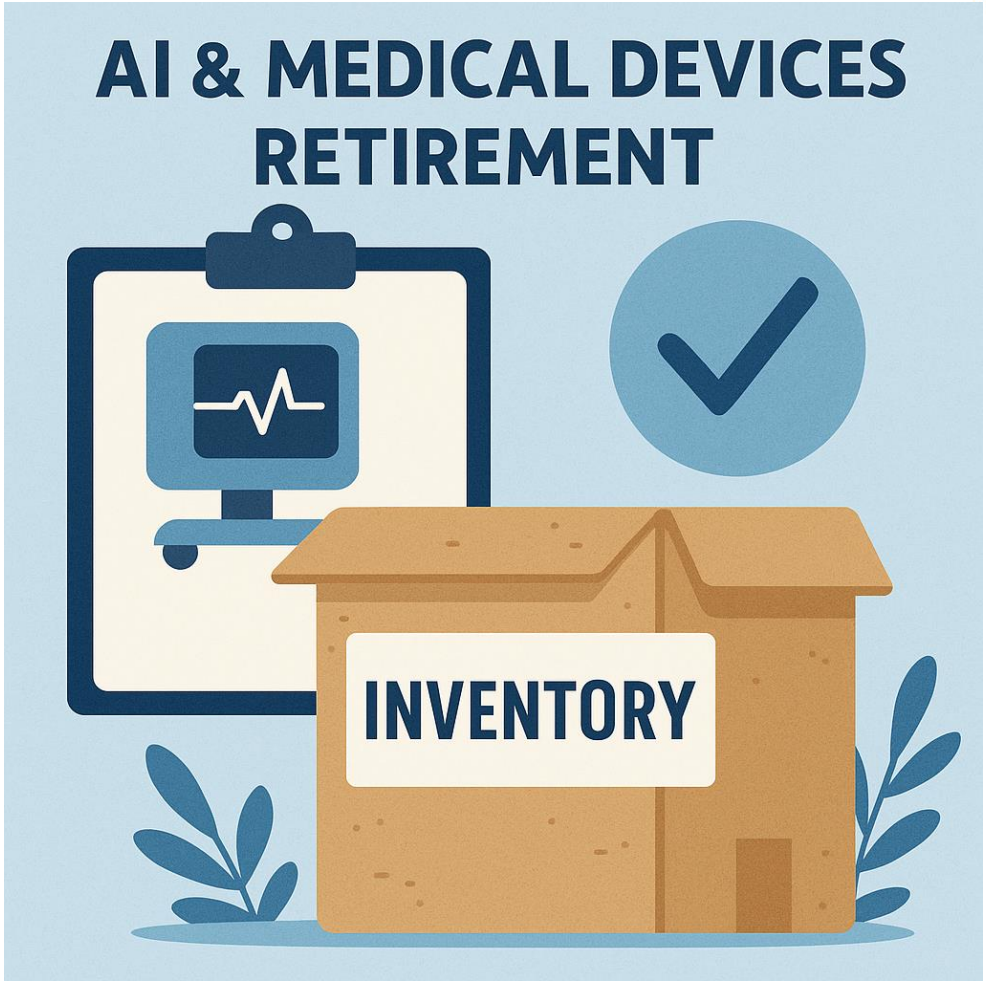
- Purpose: plan inventory shifts, predict costs, and ensure technology continuity.
- Tools :
  - AHA equipment lifetime estimates → basis for depreciation & lifespan.
  - Total cost of maintenance over device life.
  - Failure frequency & downtime history.
  - Availability of newer technology.
- **Approach: data-driven decisions using repair records + lifetime predictions.**



Ai & Medical Devices Retirement ! [The new trend]

- AI models can analyze historical maintenance data to predict when a device is approaching the end of its useful life.

- This helps hospitals make smarter decisions about retiring, replacing, or upgrading equipment.



De-Installation & Disposition

- When: device no longer repairable, outdated, or replaced.
- Process:
  - De-installation (may require 3rd party for large/regulated systems).
  - Remove all PHI/PII from device hard drives.
    - PHI = Protected Health Information
    - PII = Personally Identifiable Information
- Final outcomes:
  - Disposal
  - Resale
  - Reuse within health system
  - Donation to charities.

Think with me !? 🤔

- Why is it important to view a medical device as having a full life cycle—from need identification all the way to final disposal—instead of seeing it only as equipment we buy and use?
- How do preventive maintenance and reactive maintenance differ, and can you give an example of a device where each type would be most appropriate?
- If you were a clinical engineer, what key information would you check before deciding that a device should be retired or replaced?


جامعة الأزهر - غزة
1412-1991 - ١٤١٢-١٩٩١
AL-AZHAR UNIVERSITY - GAZA

Thank you

Abbreviation	Full Meaning
FDA	Food and Drug Administration
ICU	Intensive care unit
IMC	Intermediate care unit
MEMP	Medical equipment management plan
MI	Myocardial infarction
MRI	Magnetic resonance imaging
NFPA	National Fire Protection Association
NICU	Neonatal intensive care unit
PCA	Patient-controlled analgesia
PET	Positron emission tomography
PHI	Protected health information
PICU	Pediatric intensive care unit
RFI	Request for Information
RFP	Request for Proposal